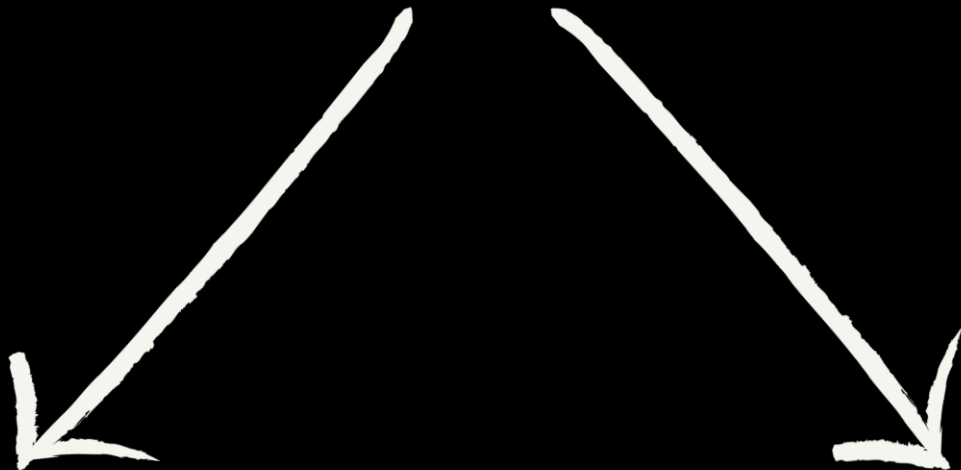


POWER PLANTS

Sources of Energy



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graph TD; A[Sources of Energy] --> B[Conventional]; A --> C[Non-Conventional];
```

Conventional

*Conventional Sources of Energy are also known as **non-renewable** sources of energy and are available in limited quantity apart from hydro-electric power.*

eg: Coal, Oil and Natural gas, Electricity(thermal, hydroelectric,nuclear).

Non-Conventional

*Non-Conventional sources of energy are called **renewable** sources, which can be easily replenished by natural processes.*

eg: Solar energy, Wind energy, Tidal energy and Bioenergy.

Hydroelectric Power

Hydroelectricity refers to the generation of electrical power by the use of hydropower.

Hydroelectric power is the electricity produced from generators driven by turbines that convert the potential energy of falling or fast-flowing water into mechanical energy.



Srisailem reservoir,
Telangana



Principle of Hydroelectric Power Generation

1. The energy of the naturally flowing water in high rivers is stored behind the dams as potential energy.

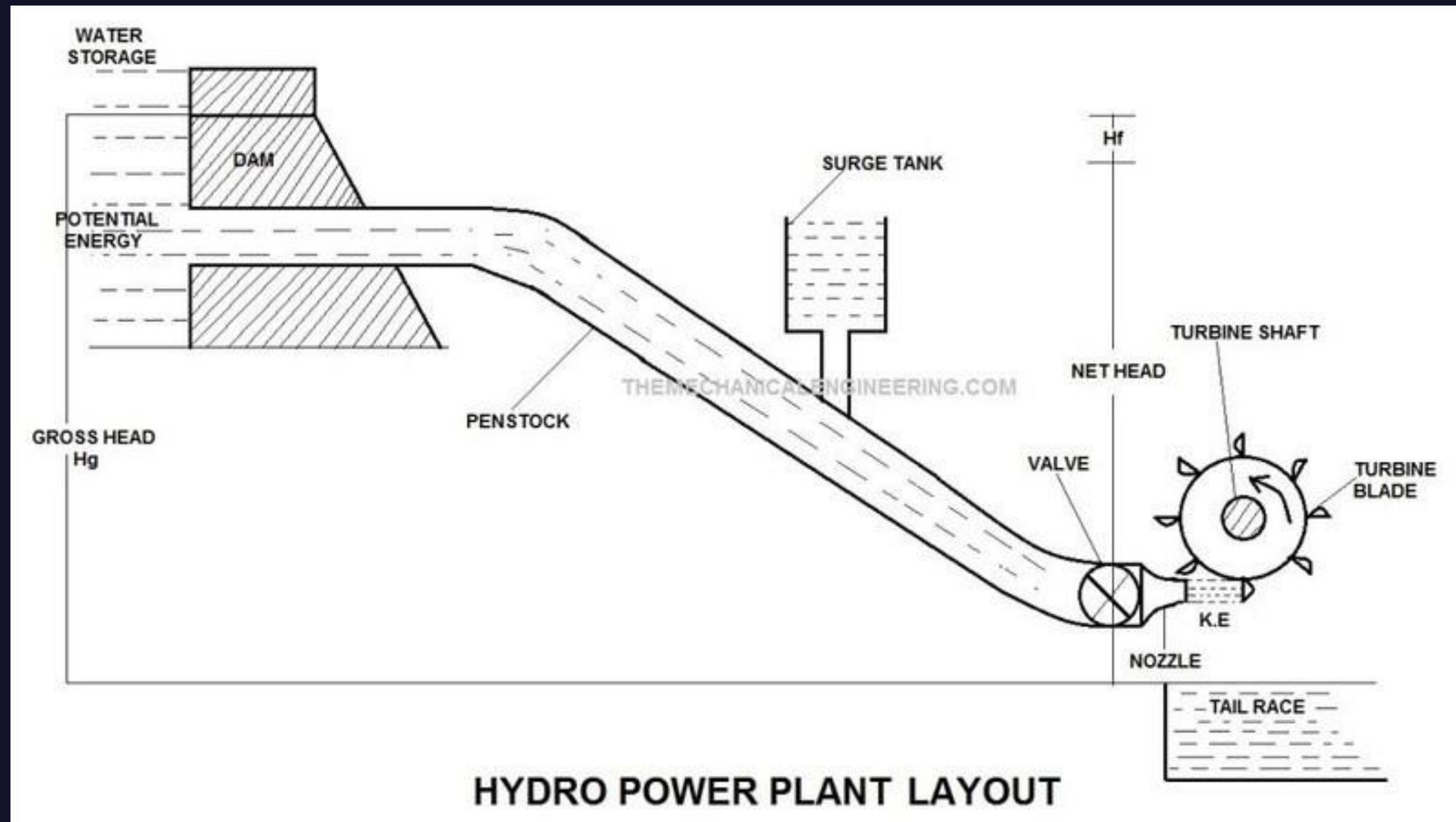
2. When this water fall on the big water wheels called turbines, the wheels starts rotating in a very high speed.

3. These processes in turn results in the rotation of armature of the generator and leads to production of electricity.

**Potential Energy → Kinetic Energy → Mechanical Energy →
Electrical Energy**

Hydroelectric Power Plant

Hydro Power Plant is an electricity-producing plant in which the water is an essential fuel, the potential energy is being converted into kinetic energy and kinetic energy is further converted into mechanical and into electrical energy with the help of a turbine and motor.



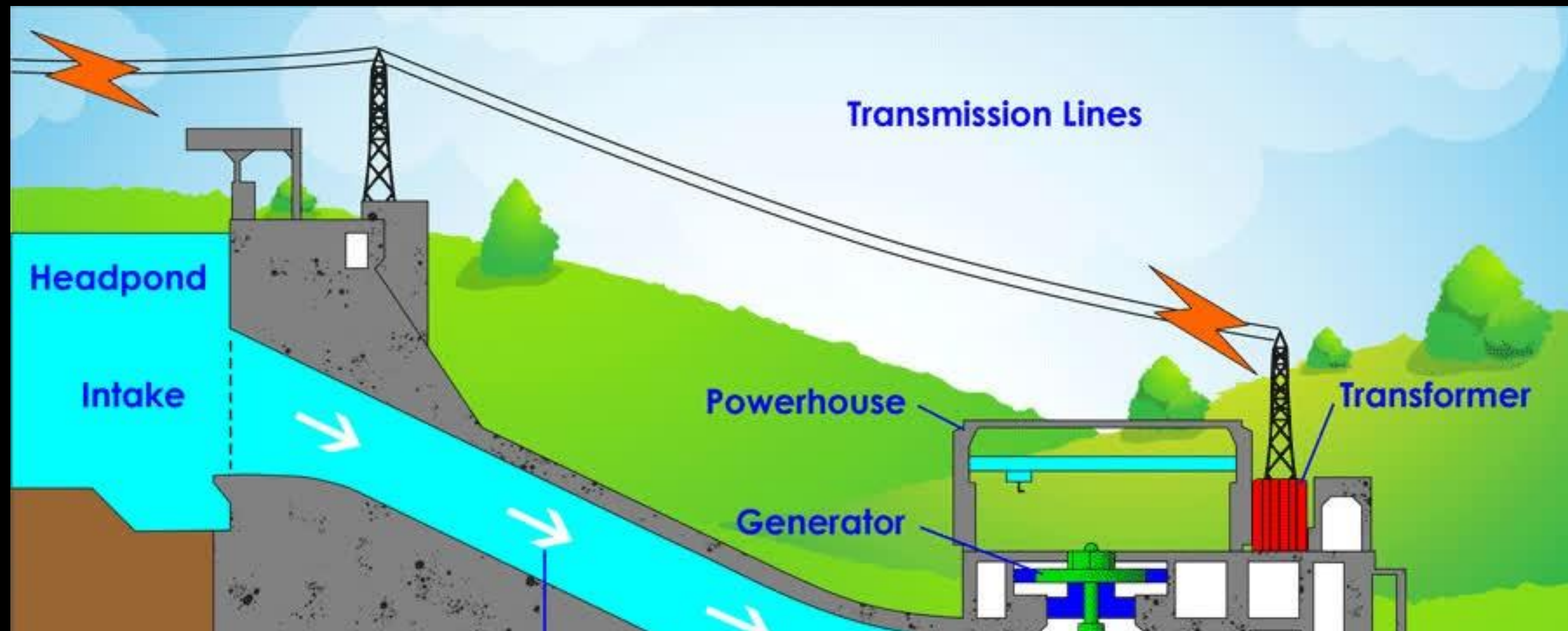
Hydroelectric Power Plant Layout

The major parts of a hydroelectric power plant are:

1. Headpond or Reservoir
2. Control gate
3. Penstock
4. Turbine
5. Draft Tube
6. Tail Race
7. Transmission Line
8. Generator
9. Transformer
10. PowerHouse

HEADPOND/RESERVOIR

There is one reservoir which is having a large area in which a huge amount of water is being stored here. So the energy here is in the form of Potential energy.

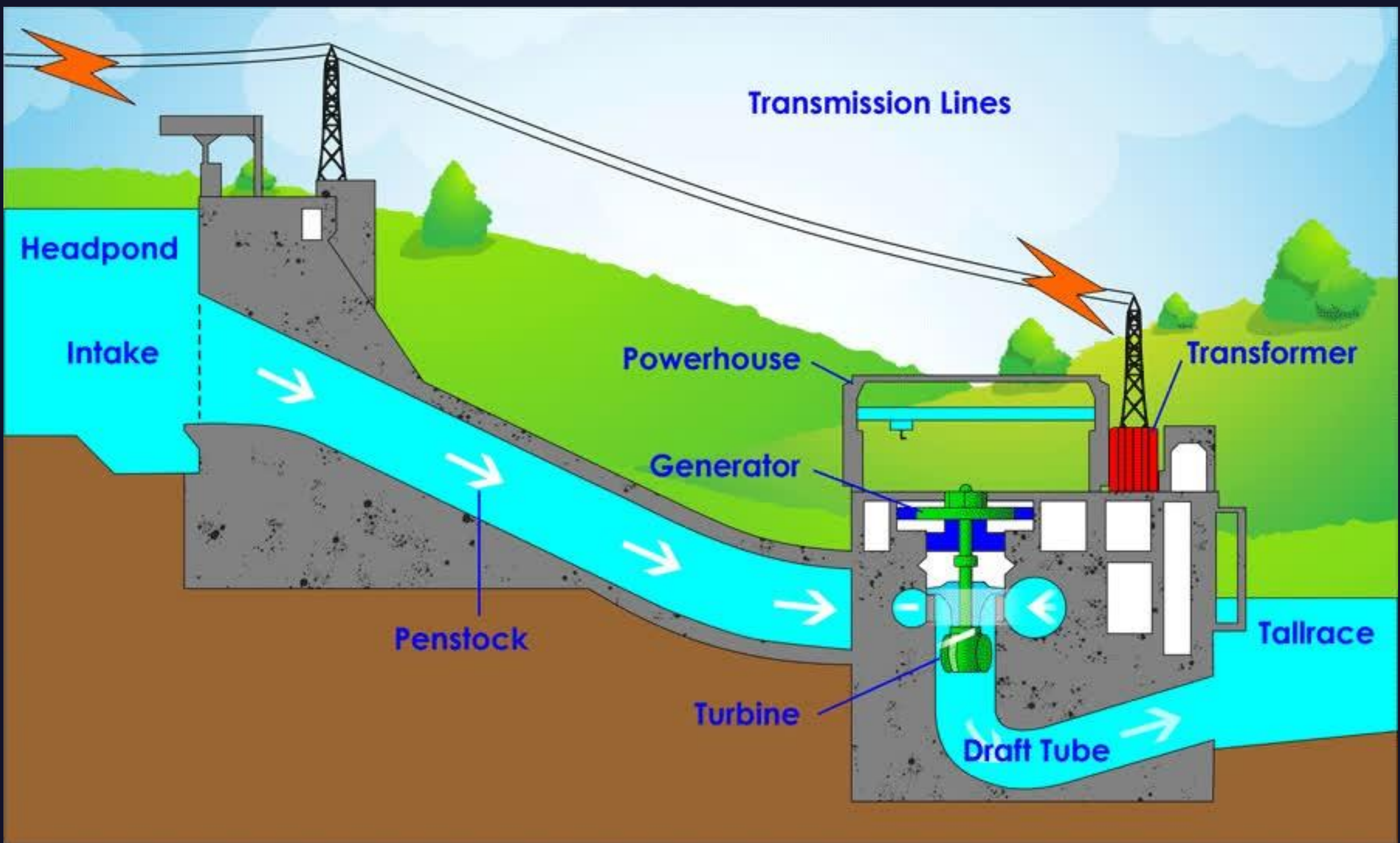


CONTROL GATE

There are having multiple control gates in a single hydro power plant. The work of control gate is to regulate the flow of water. When the control gate is fully opened the speed of water flowing is maximum.

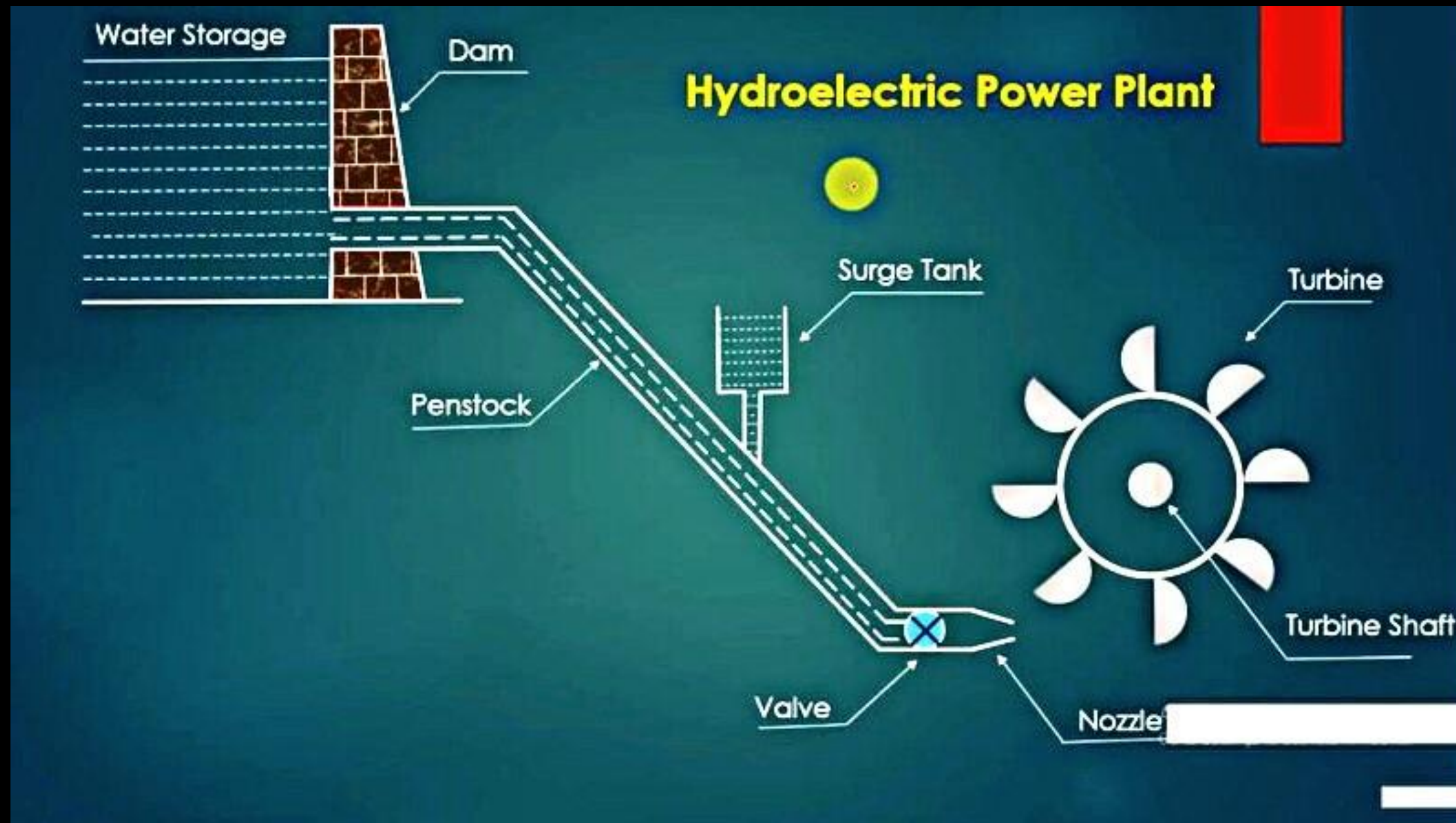
PENSTOCK

The penstock is also called Pipe. The water stored at the dam or head pond is being released by the control gate, the water starts moving to the turbine. A penstock is an enclosed pipe that delivers water to hydro turbines. Inorder to tolerate the pressure produced by the speed of water due to gravitational force, the material of the penstock is made by hard steel generally.



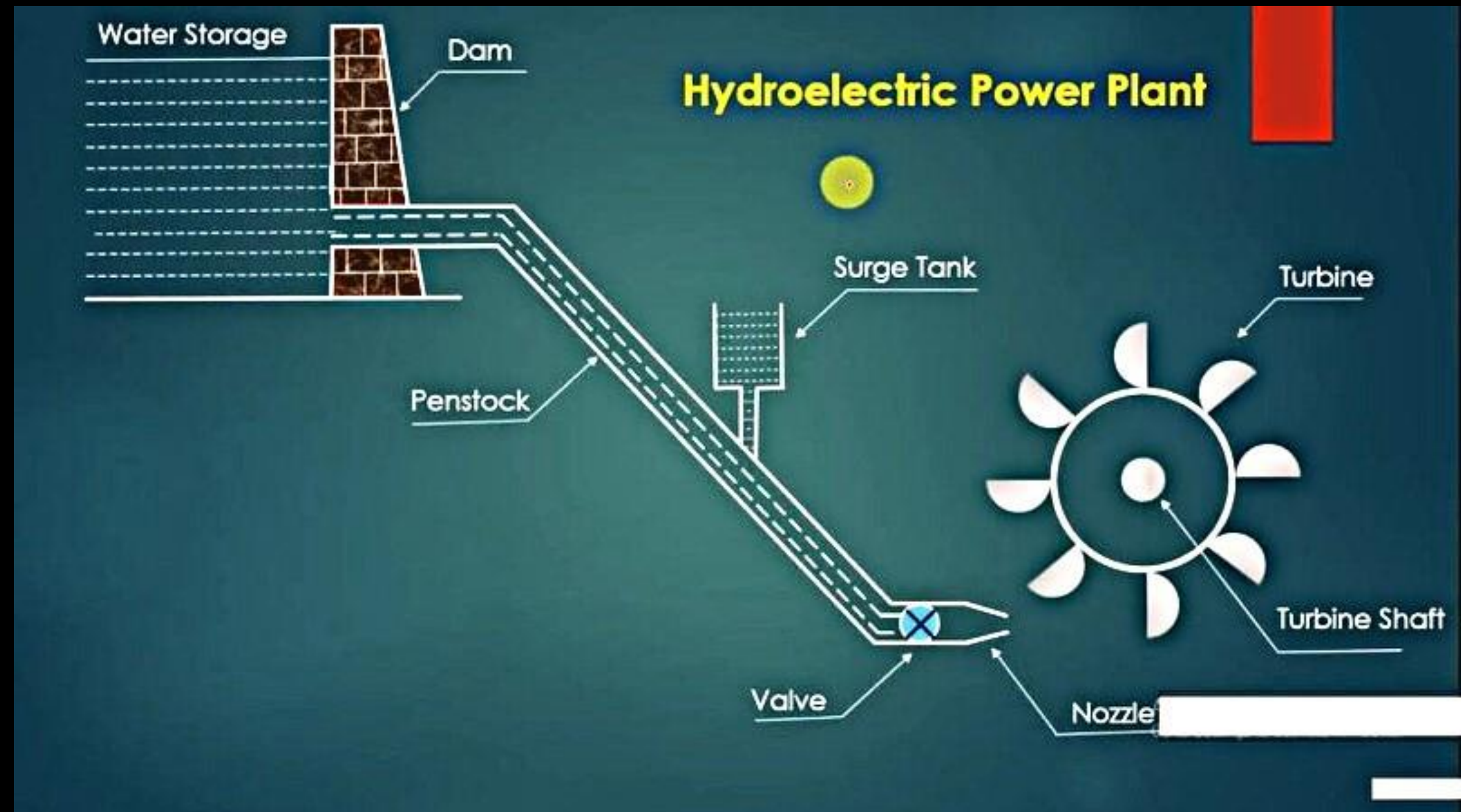
VALVE AND NOZZLE

The valve work is similar to the control gate and Nozzle regulates the rate of flow and direction of the stream of water.



SURGE TANK

Surge tank is an additional component which is used to accumulate the water which is in pipe when we want to close the turbine working. Or you can say it is used for avoiding the pipe burst.

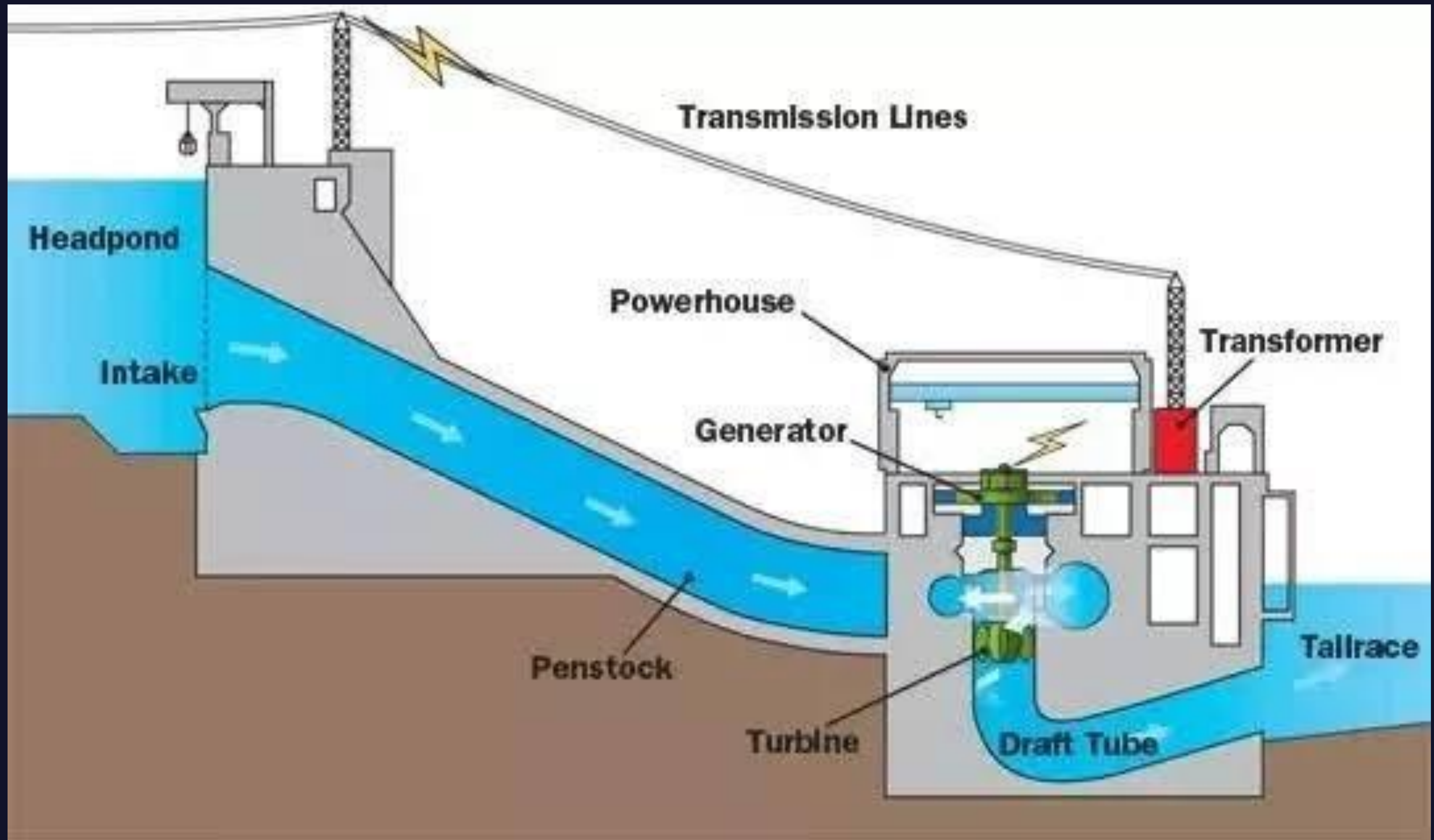


TURBINE

Turbine is a device which is used for generation of electricity. The high kinetic energy holding water comes through the penstock to the nozzle and strikes the turbine blades. The turbine blades start rotating which leads to the rotation of armature in the generator. Hence the kinetic energy turns into rotational energy, which here can also be called mechanical energy.

DRAFT TUBE

Draft tube is mechanical component which is used for enlarging the area of pipe for sending maximum fluid to the other side. It brings the pressure of the water exiting the turbine back to atmospheric pressure levels by reducing the velocity of the water at the tailrace.



GENERATOR

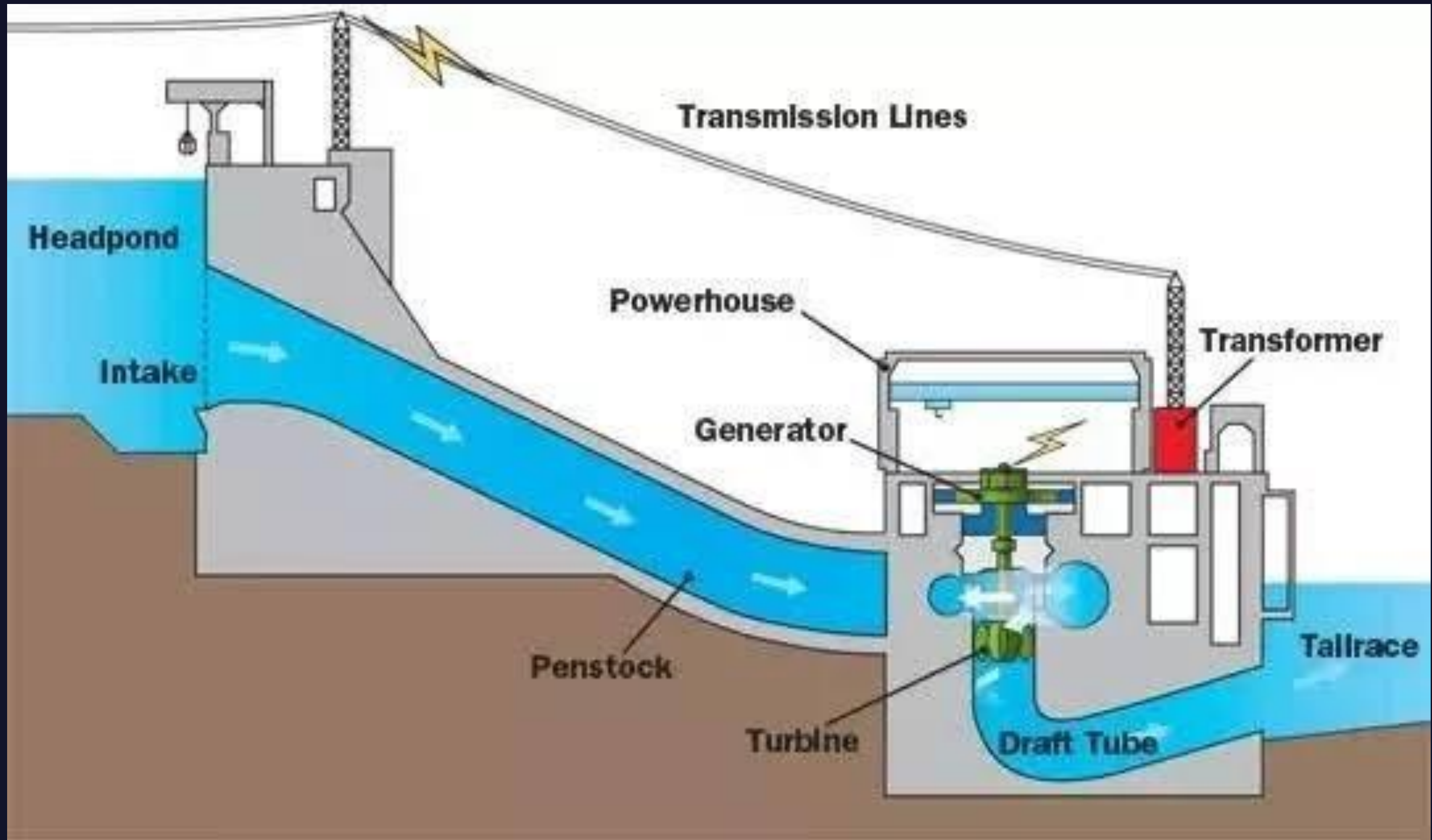
The turbines in turn drive generators, which convert the turbine's mechanical energy into electricity

TRANSFORMER

Transformers are then used to convert the alternating voltage suitable for the generators to a higher voltage suitable for long-distance transmission. Step up transformers are used here.

POWER HOUSE

The structure that houses the turbines and generators, and into which the pipes or penstocks feed, is called the powerhouse.



TRANSMISSION LINES

The transmission line carries power from the power house and transfers or supplies from one source to another.

TAIL RACE

The tail race, containing tail water, is a channel that carries water away from a hydroelectric plant or water wheel. The water in this channel has already been used to rotate the turbine and generate electricity. This water eventually joins the natural flow of river.

ADVANTAGES

- ✓ **Low operating and maintenance cost when compared to thermal and nuclear plants.**
- ✓ **Cost of power generation is low and zero fuelling.**
- ✓ **High life expectancy**
- ✓ **No production of smoke or ash, hence zero pollution.**
- ✓ **Efficiency stays constant with age.**
- ✓ **Helps in irrigation and flood control.**

DISADVANTAGES

- ✓ **Capital cost of construction is high when compared to others.**
- ✓ **It demands large area and depends on weather conditions.**
- ✓ **Long transmission lines are required as the plants are located in hilly areas.**
- ✓ **Loss of power transmission and dry year affects the power generation seriously.**